NUTRITION 10

Malnutrition plays an important role in the health and welfare of children and women in Ghana. Poor nutrition results in morbidity, mortality, poor education, and fewer opportunities for economic development. Poor education, low socio-economic status, and high fertility are factors that may influence the nutritional status of an individual. In Ghana, children under five years and women of reproductive age are the most vulnerable. Economic evidence also suggests that life expectancy is directly related to poverty and nutrition (Sachs, 1999). Adequate food and sound nutrition are essential to good health. They are crucial not only for human survival, but also for prevention of and recovery from illness.

The 2003 GDHS collected data from respondents in order to evaluate the nutritional status of women and young children. For infants and young children, this included information on breastfeeding and complementary feeding. For micronutrients like iron, vitamin A, and iodine, information was collected on intake levels from supplementation and food. Anthropometric measurements (height and weight) were taken for women 15-49 years and children under age five to determine their nutritional status.

10.1 BREASTFEEDING

Appropriate feeding practices are of fundamental importance for the survival, growth, development, health, and nutrition of infants and children and for the well being of mothers. Feeding practices are one of the underlying determinants of children's nutritional status, which in turn influence the risk of illness and ultimately death. Breastfeeding benefits depend on the length of time a child is breastfed, and the frequency and intensity of breastfeeding. Breastfeeding also affects the mother in other ways. The physiological suppression of fertility as a result of intensive breastfeeding influences the length of the interval between pregnancies.

10.1.1 Initiation of Breastfeeding

Breastfeeding is sufficient and beneficial for infant nutrition in the first six months of life. Early initiation of breastfeeding (breastfeeding within one hour) facilitates the newborn's innate sucking reflex, which helps to stimulate breast milk production and provides all of the nutritional requirements of a young infant (Righard and Alade, 1990). The high concentration of antibodies in colostrum, the first yellowish, highly nutritious milk that is present right after delivery, protects the child from infection before the child's immune system has matured. Early initiation also encourages the bond between mother and baby and helps to maintain the baby's body temperature. Breastfeeding also helps the uterus to retract, hence reducing postpartum blood loss of the mother. Prelacteal feeding (giving something other than breast milk in the first three days of life) is generally discouraged since it may inhibit breastfeeding and expose the newborn infant to illness.

The Baby Friendly Hospital Initiative (BFHI) started in health facilities in Ghana in 1993. This may have had an impact on breastfeeding initiation and the giving of prelacteal feeds among children of mothers assisted at delivery by medically trained health professionals and those delivered in a health facility.

Table 10.1 shows the percentage of children born in the five years preceding the survey who were ever breastfed and among children ever breastfed, the proportion who started breastfeeding within one

Table 10.1 Initial breastfeeding

Percentage of children born in the five years preceding the survey who were ever breastfed, and among children ever breastfed, percentage who started breastfeeding within one hour and within one day of birth and percentage who received a prelacteal feed, by background characteristics, Ghana 2003

	All children		Children ever breastfed			
Background characteristic	Percentage ever breastfed	Number of children	Percentage who started breastfeeding within 1 hour of birth	Percentage who started breastfeeding within 1 day of birth[1]	Percentage who received a prelacteal feed[2]	Number of children ever breastfed
Sex						
Male	96.9	1,841	46.8	76.0	18.6	1,784
Female	97.2	1,798	45.8	74.4	20.6	1,748
Residence						
Urban	97.8	1,204	47.2	76.7	14.3	1,178
Rural	96.7	2,435	45.9	74.5	22.2	2,354
Region						
Western	96.9	367	35.7	66.9	29.2	356
Central	97.9	304	14.2	82.6	11.1	298
Greater Accra	96.7	390	45.7	68.1	12.8	377
Volta	97.9	298	47.1	79.3	12.3	292
Eastern	97.0	362	39.9	76.2	26.2	351
Ashanti	95.6	685	51.8	74.6	20.4	655
Brong Ahafo	96.9	401	54.3	79.3	24.5	388
Northern	97.9	500	54.1	70.5	24.8	489
Upper East	98.4	215	86.3	92.9	8.6	212
Upper West	97.1	118	17.9	69.7	8.6	114
Mother's education						
No education	97.7	1,466	48.3	72.7	21.0	1,433
Primary	95.2	843	44.2	74.3	20.3	803
Middle/JSS	97.3	1,139	45.3	78.6	16.9	1,108
Secondary+	98.2	191	46.4	79.0	21.4	187
Assistance at delivery						
Health professional[3]	96.5	1,713	51.4	80.4	14.1	1,652
Traditional birth attendant	97.4	1,127	38.7	71.1	24.5	1,097
Other	97.7	696	50.0	73.2	25.0	680
No one	98.6	78	26.5	63.6	26.4	77
Place of delivery						
Health facility	96.5	1,661	51.6	80.4	13.8	1,602
At home	97.5	1,942	42.5	71.8	24.6	1,894
Wealth quintile						
Lowest	97.6	941	46.6	71.7	24.0	919
Second	97.0	809	47.1	78.0	23.8	785
Middle	95.7	721	44.1	72.3	18.5	690
Fourth	97.9	617	46.0	76.9	16.7	604
Highest	96.9	551	48.1	79.3	10.4	534
Total	97.0	3,639	46.3	75.2	19.6	3,532

Note: Table is based on all births whether the children are living or dead at the time of interview. Total includes 27 children with missing information on assistance at delivery, 24 persons with missing information on place of delivery, and 14 persons with 'other' place of delivery, who are not shown separately. An asterisk indicates that a figure is based on fewer than 25 unweighted cases and has been suppressed.

[1] Includes children who started breastfeeding within one hour of birth

[2] Children given something other than breast milk during the first three days of life before the mother started breastfeeding regularly

[3] Doctor, nurse/midwife, or auxiliary midwife

hour and within one day of birth, and those who received a prelacteal feed. The data indicate that almost all (97 percent) Ghanaian children are breastfed for some period of time. Forty-six percent of infants were put to the breast within one hour of birth, and 75 percent started breastfeeding within the first day. The data from 2003 can be compared with similar data collected five years ago. The data show that over the past five years, there was little difference in the percentage of children ever breastfed. However, the percentage breastfed within one hour of birth and one day of birth for children born in the five years preceding the survey (46 and 75 percent, respectively) is noticeably higher than the 1998 levels (25 and 54 percent, respectively) (GSS and MI, 1999).

There are no marked differences in the proportion of children ever breastfed by background characteristics. Children of mothers assisted at delivery by medically trained health professionals (doctors, nurse/midwives, or auxiliary nurses) and children delivered in a health facility are more likely to be breastfed immediately after birth or within one day of birth.

Initiation of breastfeeding varies among regions. The proportion of infants that are breastfed within one hour of birth ranges from 14 percent in the Central Region to 86 percent in the Upper East Region. The Western Region has the lowest percentage of children who started breastfeeding within one day of birth (67 percent), while the Upper East Region has the highest (93 percent).

Prelacteal feeding, something other than breast milk given to newborns prior to the regular flow of breastmilk, is not widely practised in Ghana. Only 20 percent of children born in the five years preceding the survey received a prelacteal feed. Prelacteal feeding is more widely practiced in rural areas (22 percent) than urban areas (14 percent). The Western Region (29 percent) has the highest reported percentage of prelacteal feeding. Children of mothers assisted at delivery by medically trained health professionals have a lower reported rate of receiving prelacteal feeds (14 percent). Women who delivered at home have higher reported rates of prelacteal feeding (25 percent) than those who delivered in a health facility (14 percent). The practice decreases from 24 percent among children of women in the lowest wealth quintile to 10 percent among children of women in the highest wealth quintile.

10.1.2 Age Pattern of Breastfeeding

Breast milk is safe, convenient, uncontaminated, and contains all the nutrients needed by the baby in the first six months of life. In Ghana, all women are encouraged to breastfeed their infants exclusively for the first six months and then complement the breastfeeding with nutritious foods for at least two years. Breast milk provides protection against infection through the mother's antibodies. Supplementing breast milk before 6 months is unnecessary and is strongly discouraged because of the likelihood of contamination, the unaffordability of breast milk substitutes, and the resulting increased risk of diarrhoeal disease. The early introduction of liquids and solids reduces breast milk output because the production and release of milk is influenced by the frequency and intensity of suckling. Breastfeeding remains the best nutrition even for infants of HIV-positive mothers and mothers whose HIV status is unknown, as it provides resistance to opportunistic diseases.

Table 10.2 shows the percent distribution of youngest children under three years living with the mother by breastfeeding status and percentage of children under three years using a bottle with a nipple, according to age in months. Based on information about feeding practices in the 24 hours preceding the survey, almost all children are breastfed for at least one year, with only 4 percent of children aged 12-15 months who are not breastfed. By age 16-19 months, 14 percent of children are no longer breastfeeding. Breastfeeding decreases rapidly late in the second year of life, and by 32-35 months of age, virtually all children (94 percent) are weaned.

Table 10.2 Breastfeeding status by age

Percent distribution of youngest children under three years living with the mother by breastfeeding status and percentage of children under three years using a bottle with a nipple, according to age in months, Ghana 2003

Age in months	Not breast-feeding	Exclusively breastfed	Breastfeeding and consuming: Plain water only	Water-based liquids/ juice	Other milk	Comple-mentary foods	Total	Number of children	Percentage using a bottle with a nipple[1]	Number of children
<2	1.3	62.4	31.1	1.5	1.4	2.3	100.0	92	9.4	94
2-3	0.0	64.9	22.4	0.0	7.0	5.7	100.0	88	9.7	90
4-5	0.0	38.9	19.9	1.8	7.3	32.0	100.0	127	15.1	129
6-7	0.8	14.3	27.1	2.7	3.1	52.1	100.0	137	12.9	139
8-9	0.0	3.2	17.0	1.9	2.1	75.8	100.0	102	11.1	102
10-11	1.3	3.4	8.4	0.7	0.0	86.2	100.0	131	7.5	134
12-15	4.2	0.7	7.3	0.0	1.0	86.8	100.0	258	8.9	270
16-19	14.2	0.5	5.5	0.6	0.0	79.2	100.0	224	9.1	230
20-23	33.3	0.0	2.9	0.0	0.7	63.1	100.0	181	5.4	195
24-27	70.1	0.2	0.4	1.0	0.0	28.4	100.0	202	5.3	227
28-31	82.2	0.4	0.5	0.0	0.0	16.9	100.0	158	6.4	212
32-35	93.8	0.0	0.0	0.0	0.0	6.2	100.0	146	6.0	210
<6	0.4	53.4	24.0	1.2	5.5	15.6	100.0	308	11.8	314
6-9	0.4	9.6	22.8	2.4	2.7	62.2	100.0	239	12.2	241

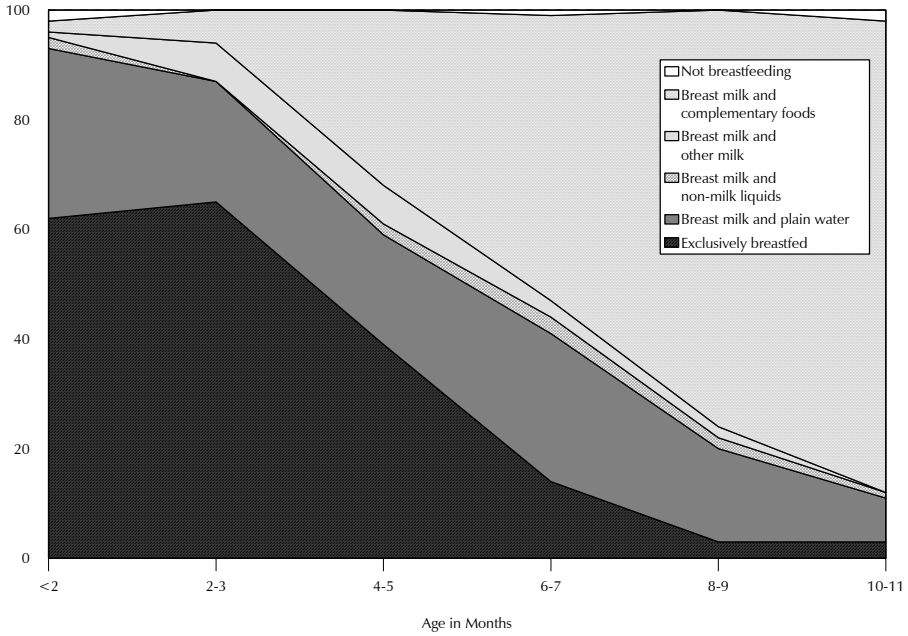
Note: Breastfeeding status refers to a "24-hour" period (yesterday and last night). Children classified as *breastfeeding and consuming plain water only* consume no supplements. The categories of not breastfeeding, exclusively breastfed, breastfeeding and consuming plain water, water-based liquids/juice, other milk, and complementary foods (solids and semi-solids) are hierarchical and mutually exclusive, and their percentages add to 100 percent. Thus children who receive breast milk and water-based liquids and who do not receive complementary foods are classified in the water-based liquid category even though they may also get plain water. Any children who get complementary food are classified in that category as long as they are breastfeeding as well.

[1] Based on all children under three years

Despite the high breastfeeding prevalence (97 percent) in Ghana, the majority of infants are not fed in compliance with the WHO/UNICEF recommendations (World Health Assembly, 2001). These recommendations call for a period of exclusive breastfeeding for 6 months and the introduction of complementary foods after the age of 6 months. Fifty-three percent of children under 6 months of age are exclusively breastfed in Ghana. This is a slight increase over the proportion reported in the 1998 GDHS (GSS and MI, 1999).

Exclusive breastfeeding drops sharply from 65 percent at age 2-3 months to 39 percent at age 4-5 months (Figure 10.1). Six percent of children age 2-3 months and 32 percent of children age 4-5 months are receiving complementary foods in addition to breast milk. This indicates that there are many infants who are at risk of being exposed to bacterial contamination and poor quality foods, even if they started out well with early initiation of breastfeeding. The duration of paid maternity leave in Ghana is 12 weeks; hence, a majority of women return to work when their children are three months old. This may account for the sharp decline in exclusive breastfeeding rates between 2-3 months and 4-5 months.

Figure 10.1 **Breastfeeding Practices by Age, Ghana 2003**



The use of a feeding bottle with a nipple is discouraged in Ghana. HIV-positive mothers are told not to use feeding bottles even when they choose to use formula as their feeding option. The use of a bottle with a nipple, regardless of the contents (formula or any other liquid), requires hygienic handling. As a result of inadequate and insufficient cleaning and ease of recontamination after cleaning, the nipple may house disease-causing agents transferable to the baby. Table 10.2 indicates that 12 percent of children under six months and the same proportion of children age 6-9 months are given a feeding bottle with a nipple. Bottle-feeding reaches its peak (15 percent) at age 4-5 months. The percentage of children who are bottle-fed declines to 5 percent by the age of two years. It however increases slightly after that to 6 percent by 28 months. It is assumed that by two years most children are eating solid foods, which does not require feeding by bottle. The percentage of young children bottle-fed has declined markedly over the past five years. For example, bottle-feeding at age 4-5 months has declined from 26 percent in 1998 (GSS and MI, 1999) to 15 percent in 2003.

Table 10.3 presents information on the median duration of any breastfeeding, exclusive breastfeeding, and predominant breastfeeding among children born in the three years preceding the survey, the percentage of breastfeeding children under six months living with the mother who were breastfed six or more times in the 24 hours preceding the survey, and the mean number of feeds (day and night), by background characteristics.

The median duration of any breastfeeding in Ghana is 23 months. Regional differences in breastfeeding prevalence are minimal, with the longest duration being 28 months in the Northern Region and the lowest (19 months) in Greater Accra. The median duration of exclusive breastfeeeding is 2 months and the median duration of predominant breastfeeding is 5 months. Children are considered predominantly breastfed when they are either exclusively breastfed or receive breast milk and plain water, water-based liquids, and/or juice only (excluding other milk and solids).

Frequent breastfeeding improves the production of breast milk. It is also a benefit for some mothers who breastfeed exclusively as it delays the return of fertility.

Table 10.3 indicates that almost all breastfed children under 6 months (97 percent) are breastfed six or more times in the past 24 hours, with an average of 7 daytime feeds and 5 nighttime feeds.

Table 10.3 Median duration and frequency of breastfeeding

Median duration of any breastfeeding, exclusive breastfeeding, and predominant breastfeeding among children born in the three years preceding the survey, percentage of breastfeeding children under six months living with the mother who were breastfed six or more times in the 24 hours preceding the survey, and mean number of feeds (day/night), by background characteristics, Ghana 2003

	Median duration (months) of breastfeeding[1]				Breastfeeding children under six months[2]			
Background characteristic	Any breast-feeding	Exclusive breastfeeding	Predominant breastfeeding[3]	Number of children	Percentage breastfed 6+ times in last 24 hours	Mean number of day feeds	Mean number of night feeds	Number of chil-dren
Sex								
Male	22.8	2.8	5.1	1,090	96.0	7.7	4.6	168
Female	22.3	1.2	5.1	1,085	97.9	7.1	4.4	143
Residence								
Urban	20.4	4.1	5.0	732	99.3	8.1	4.3	97
Rural	23.3	1.4	5.1	1,442	95.8	7.1	4.6	214
Region								
Western	20.2	0.6	2.4	215	(100.0)	(7.8)	(4.9)	34
Central	21.3	0.5	6.3	184	*	*	*	22
Greater Accra	19.3	5.7	6.9	229	(100.0)	(7.7)	(4.9)	28
Volta	22.9	3.2	4.1	184	*	*	*	27
Eastern	21.2	2.9	3.4	220	*	*	*	23
Ashanti	20.6	1.9	3.2	407	(96.3)	(7.8)	(4.4)	59
Brong Ahafo	22.6	3.5	6.2	240	(100.0)	(9.0)	(4.3)	39
Northern	28.2	0.7	7.4	297	92.5	6.2	3.5	47
Upper East	26.4	1.4	8.3	129	(97.2)	(4.8)	(4.7)	20
Upper West	27.4	5.1	7.0	70	(84.3)	(10.0)	(4.7)	13
Mother's education								
No education	25.3	2.0	6.4	863	94.5	7.2	4.3	139
Primary	21.6	1.9	4.1	511	98.2	7.4	5.0	66
Middle/JSS	20.8	2.6	4.2	676	99.0	7.6	4.5	96
Secondary +	17.4	2.9	4.1	124	*	*	*	11
Wealth quintile								
Lowest	26.8	2.4	6.9	548	96.0	6.9	4.7	76
Second	22.8	0.7	5.2	477	94.4	7.4	4.5	81
Middle	21.6	1.8	4.6	450	97.2	7.9	4.6	78
Fourth	20.6	3.0	3.9	364	(100.0)	(7.5)	(4.3)	42
Highest	19.2	3.7	4.4	336	(100.0)	(7.8)	(4.3)	34
Total	22.5	2.3	5.1	2,175	96.9	7.4	4.5	312
Mean for all children	22.2	3.8	6.9	na	na	na	na	na

Note: Median and mean durations are based on current status. An asterisk indicates that a figure is based on fewer than 25 unweighted cases and has been suppressed. Percentages in parentheses are based on 25-49 unweighted cases.
na = Not applicable
[1] It is assumed that non-last-born children or last born child not living with the mother are not currently breastfeeding
[2] Excludes children who do not have a valid answer on the number of times breastfed
[3] Either exclusively breastfed or received breast milk and plain water, water-based liquids, and/or juice only (excludes other milk)

Although the median duration of breastfeeding is slightly longer in the rural areas (23 months) than in the urban areas (20 months), exclusive breastfeeding is somewhat shorter in rural areas (1 month) than in urban areas (4 months). Exclusive breastfeeding rates increase with increases in mother's level of education and wealth quintile.

10.2 COMPLEMENTARY FEEDING

10.2.1 Types of Complementary Foods

In line with the WHO/UNICEF global strategy on infant and young child feeding, the Ghana Health Service recommends 6 months as the optimal age to introduce complementary foods. This is because after 6 months of age, breast milk alone is not sufficient to meet all the nutritional requirements of the infant. The period after 6 months of age is a crucial time for children because they are being introduced to the family diet, and a number of issues come into force. These include income, hygiene, general care, and choice of complementary foods. The spread of HIV/AIDS has created another challenge for breastfeeding.

Ghana has in place legislation under the Food and Drugs Law to control the marketing of breast milk substitutes by baby food manufacturers. The legislative instrument Breastfeeding Promotions Regulation 2000 (L.I.1667) went into effect in May 2000. The purpose is to prevent the aggressive marketing of breast milk substitutes, hence protecting breastfeeding practices.

Table 10.4 shows the percentage of youngest children under three years of age living with the mother who consumed specific foods in the day or night preceding the interview, by breastfeeding status and age. It is important to note that the categories presented in Table 10.4 are not mutually exclusive. The child who consumed milk could also have consumed semi-solid foods. The data show that among breastfeeding infants in Ghana, very few receive infant formula. Forty-seven percent of breastfeeding children age 4-5 months are introduced to solid or semi-solid foods and 30 percent are introduced to foods made from grains.

Table 10.4 also shows that at the age of 6-9 months, more than 70 percent of breastfeeding infants received solid foods in the 24 hours before the survey. The majority (53 percent) consumed food made from grains, 29 percent received fruits and vegetables, and 21 percent received animal products (in addition to breast milk). Foods rich in vitamin A were consumed by only 24 percent of breastfeeding infants 6-9 months, and fats by only 8 percent. The findings indicate that, in terms of nutrition, the food mixtures given during this transition period are somewhat limited. By 10-11 months, almost all (91 percent) infants are receiving solids foods. This is a marked improvement over the 1998 GDHS figure of 73 percent for this age group (GSS and MI, 1999). By that same age, 10-11 months, a larger proportion are consuming grains (76 percent), fruits and vegetables (53 percent), and foods rich in vitamin A (46 percent). Although animal products (a major source of iron and vitamin A) are consumed by 46 percent of children in this age group, consumption of animal products is not as prevalent as the other foods. By 20-23 months, grains (84 percent), fruits and vegetables (62 percent), and foods rich in vitamin A (53 percent) are consumed by the majority of breastfeeding children; consumption of animal products increases slightly to 58 percent.

Few children under two years of age are not breastfed. For non-breastfeeding children about two years of age, the rates of consumption of food are about the same as for breastfeeding children; however, these children do not have the added nutritional benefit of breast milk, and very few receive other milk, but these children do consume more meat than breastfed children.

Table 10.4 Foods consumed by children in the day or night preceding the interview

Percentage of youngest children under three years of age living with the mother who consumed specific foods in the day or night preceding the interview, by breastfeeding status and age, Ghana 2003

Age in months	Infant formula	Other milk/ cheese/ yoghurt	Other liquids[1]	Solid/semi-solid foods						Fruits and vegetables rich in vitamin A[3]	Any solid or semi-solid food	Number of children
				Food made from grains	Fruits/ vegetables[2]	Food made from roots/ tubers	Food made from legumes	Meat/ fish/ shellfish/ poultry/ eggs	Food made with oil/ fat/ butter			
BREASTFEEDING CHILDREN												
<2	1.4	1.4	2.9	2.3	0.4	0.4	0.0	0.0	0.0	0.4	3.3	91
2-3	1.7	7.8	1.7	5.7	1.1	0.0	0.0	0.0	0.0	1.1	15.6	88
4-5	6.3	9.9	7.1	30.3	3.6	1.7	0.9	0.9	0.0	2.7	46.5	127
6-7	9.9	13.1	15.7	42.9	13.5	6.9	6.7	14.2	6.0	11.6	62.0	136
8-9	12.9	19.4	32.8	66.8	49.5	19.6	14.3	29.9	10.9	39.5	83.2	102
10-11	6.7	13.8	25.2	75.7	53.1	30.4	13.5	46.0	17.1	45.7	90.8	129
12-15	6.3	13.9	26.8	78.9	60.9	33.0	20.3	58.4	19.8	50.2	96.0	247
16-19	4.8	17.0	34.3	79.5	62.2	30.6	22.2	58.1	23.7	48.6	95.7	192
20-23	6.2	14.8	36.9	84.3	61.7	39.6	26.6	58.4	25.3	52.7	99.0	121
24-35	3.1	7.1	34.1	86.5	57.3	39.0	23.9	50.7	23.6	47.5	96.3	98
<6	3.5	6.8	4.3	14.9	1.9	0.9	0.4	0.4	0.0	1.6	24.8	306
6-9	11.2	15.8	23.0	53.1	28.9	12.3	10.0	20.9	8.1	23.5	71.1	238
NON-BREASTFEEDING CHILDREN												
16-19	(17.5)	(37.2)	(39.6)	(95.2)	(72.5)	(53.7)	(30.1)	(83.0)	(42.8)	(45.6)	(100.0)	32
20-23	7.3	25.9	37.4	93.2	70.3	47.4	31.4	78.6	43.9	60.4	100.0	60
24-35	4.5	19.2	43.0	89.2	73.4	49.5	27.1	71.8	28.7	62.0	99.0	409

Note: Breastfeeding status and food consumed refer to a "24-hour" period (yesterday and last night). Percentages in parentheses are based on 25-49 unweighted cases.

[1] Does not include plain water

[2] Includes fruits and vegetables rich in vitamin A

[3] Includes pumpkin, red or yellow yams or squash, carrots, red sweet potatoes, green leafy vegetables, mangoes, papayas, and other locally grown fruits and vegetables that are rich in vitamin A

10.2.2 Frequency of Foods Consumed by Children

Table 10.5 and Figure 10.2 present the mean number of times specific foods were consumed in the day or night preceding the interview by youngest children under three years of age living with the mother, according to breastfeeding status and age. Infants and young children eat small meals and, therefore, frequent meals are necessary to provide them with the required nutrients. It is recommended that children age 6-8 months eat a minimum of 2-3 meals and snacks per day in addition to breast milk. For children over 8 months of age, 3-5 meals should be consumed by breastfed children (WHO, 1998). The number of meals required is based on the energy density of the foods being fed. Consuming an appropriate variety of foods is essential for the child's nutrition.

Table 10.5 shows that on average foods made from grains are given to breastfeeding children only once a day from age 6-9 months, which is the best time for introducing complementary foods. Foods containing grain include flour made from maize, millet, or sorghum, which is used to make a fermented or

Table 10.5 Frequency of foods consumed by children in the day or night preceding the interview

Mean number of times specific foods were consumed in the day or night preceding the interview by youngest children under three years of age living with the mother, according to breastfeeding status and age, Ghana 2003

Age in months	Infant formula	Other milk/ cheese/ yoghurt	Other liquids[1]	Solid/semi-solid foods: Food made from grains	Fruits/ vegetables[2]	Food made from roots/ tubers	Food made from legumes	Meat/ fish/ shellfish/ poultry/ eggs	Food made with oil/ fat/ butter	Fruits and vegetables rich in vitamin A[3]	Number of children
BREASTFEEDING CHILDREN											
<2	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	91
2-3	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	88
4-5	0.1	0.2	0.1	0.5	0.1	0.0	0.0	0.0	0.0	0.1	127
6-7	0.1	0.1	0.2	0.8	0.3	0.1	0.1	0.2	0.1	0.2	136
8-9	0.3	0.4	0.6	1.2	1.1	0.2	0.2	0.5	0.2	0.7	102
10-11	0.3	0.4	0.5	1.5	1.3	0.5	0.3	0.7	0.3	0.9	129
12-15	0.1	0.2	0.4	1.4	1.3	0.4	0.3	1.0	0.3	0.9	247
16-19	0.1	0.2	0.5	1.4	1.4	0.4	0.3	1.0	0.3	0.9	192
20-23	0.1	0.3	0.7	1.8	1.5	0.5	0.3	0.9	0.3	1.1	121
24-35	0.0	0.1	0.5	1.8	1.4	0.6	0.3	0.9	0.3	1.0	98
<6	0.1	0.1	0.1	0.3	0.0	0.0	0.0	0.0	0.0	0.0	306
6-9	0.2	0.2	0.4	1.0	0.6	0.1	0.1	0.3	0.1	0.4	238
NON-BREASTFEEDING CHILDREN											
16-19	(0.3)	(0.4)	(0.6)	(2.0)	(1.7)	(0.6)	(0.3)	(1.8)	(0.5)	(1.0)	32
20-23	0.1	0.5	0.6	1.7	2.0	0.7	0.4	1.4	0.5	1.2	60
24-35	0.1	0.3	0.7	1.7	2.0	0.7	0.3	1.4	0.4	1.3	409

Note: Breastfeeding status and food consumed refer to a "24-hour" period (yesterday and last night). Percentages in parentheses are based on 25-49 unweighted cases.

[1] Does not include plain water

[2] Includes fruits and vegetables rich in vitamin A

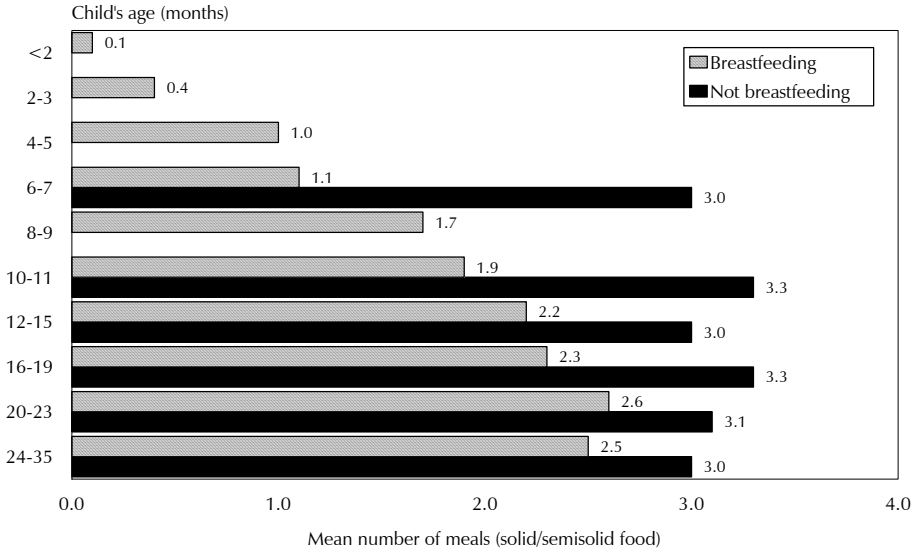
[3] Includes pumpkin, red or yellow yams or squash, carrots, red sweet potatoes, green leafy vegetables, mangoes, papayas, and other locally grown fruits and vegetables rich in vitamin A

unfermented porridge (*koko*). This food is consumed a little more than once a day through one year of age and twice a day at about two years of age for breastfed children.

At two years, non-breastfeeding children are receiving foods made from grain twice a day like breastfeeding children. Fruits and vegetables are consumed about twice a day by both breastfeeding and non-breastfeeding children at about two years. Foods enriched with oil/fat/butter—thereby increasing caloric intake—are consumed less than once a day. Animal products are consumed an average of once a day by both groups of children at two years of age.

Table 10.6 presents the mean number of days specific foods were received in the seven days preceding the interview by youngest children under three years of age, living with the mother, by breastfeeding status and age. Plain water is given almost on a daily basis. It also shows that foods most commonly given to children are those made from grains. They are given about four times a week. Animal products and green leafy vegetables are given three times a week.

Figure 10.2 **Frequency of Meals Consumed by Children under 36 Months of Age Living with Their Mother, by Breastfeeding Status, Ghana 2003**



Note: Data are not shown for groups with fewer than 25 unweighted cases. GDHS 2003

Table 10.6 Frequency of foods consumed by children in preceding seven days

Mean number of days specific foods were received in the seven days preceding the interview by youngest children under three years of age living with the mother, by breastfeeding status and age, Ghana 2003

	Liquids					Solid/semisolid foods							Fruits and vegetables rich in vitamin A			
Age in months	Plain water	Infant formula	Other milk	Fruit juice	Other liquids	Food made from grains	Food made from roots/ tubers	Fruits and vegetables not rich in vitamin A	Food made from legumes	Cheese/ yoghurt	Meat/ fish/ shellfish/ poultry/ eggs	Food made with oil/ fat/ butter	Pumpkin/ red or yellow yams/ carrots/ red sweet potatoes	Green leafy vegetables	Mango/ papaya/ other local fruits rich in vitamin A	Number of children
BREASTFEEDING CHILDREN																
<2	2.2	0.1	0.1	0.0	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	91
2-3	2.0	0.2	0.4	0.1	0.1	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	88
4-5	3.9	0.4	0.7	0.2	0.1	2.0	0.1	0.1	0.1	0.0	0.1	0.0	0.1	0.1	0.0	127
6-7	5.7	0.8	1.0	0.4	0.4	2.9	0.3	0.3	0.4	0.1	0.6	0.2	0.2	0.4	0.2	136
8-9	6.5	0.9	1.0	1.1	0.5	4.3	1.2	1.5	0.9	0.2	1.6	0.6	0.7	1.6	0.5	102
10-11	6.7	0.5	0.6	0.9	0.7	4.7	1.8	1.5	0.9	0.1	2.8	0.8	0.8	2.0	0.6	129
12-15	6.8	0.4	0.9	1.0	0.7	5.0	1.8	1.8	1.2	0.1	3.6	1.0	1.1	2.3	0.5	247
16-19	6.8	0.3	0.8	1.3	1.0	5.1	1.8	2.1	1.4	0.2	3.9	1.3	1.1	2.4	0.8	192
20-23	6.9	0.4	0.7	1.0	1.6	5.5	2.3	1.9	1.3	0.1	3.5	1.3	1.5	2.4	0.7	121
24-35	6.9	0.2	0.3	0.9	1.3	5.9	2.2	1.8	1.4	0.1	3.2	1.5	1.0	2.6	0.7	98
<6	2.9	0.2	0.5	0.1	0.1	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	306
6-9	6.1	0.8	1.0	0.7	0.5	3.5	0.6	0.9	0.6	0.1	1.1	0.4	0.4	0.9	0.3	238
Total	5.8	0.4	0.7	0.8	0.7	3.9	1.3	1.3	0.8	0.1	2.2	0.7	0.7	1.6	0.4	1,330
NON-BREASTFEEDING CHILDREN																
16-19	(6.9)	(1.3)	(2.0)	(1.4)	(0.9)	(6.0)	(2.9)	(3.2)	(1.5)	(0.2)	(5.4)	(2.3)	(1.1)	(2.6)	(1.2)	32
20-23	6.9	0.6	1.3	1.8	0.9	5.6	2.9	3.4	1.8	0.6	4.9	2.1	2.0	2.3	0.8	60
24-35	6.8	0.3	1.1	1.8	1.3	5.7	2.7	2.6	1.6	0.3	4.6	1.5	1.6	2.6	1.1	409
Total	6.8	0.4	1.2	1.8	1.2	5.7	2.7	2.7	1.6	0.3	4.6	1.7	1.6	2.5	1.0	516

Note: Breastfeeding status refers to a "24-hour" period (yesterday and last night). Percentages in parentheses are based on 25-49 unweighted cases.

10.3 MICRONUTRIENTS

Micronutrients are essential for the metabolic processes in the body and they play a major role in the nutrition and health of an individual. The 2003 GDHS collected various types of data that are useful in assessing the micronutrient status of young children and women.

10.3.1 Iodisation of Household Salt

In Ghana, a number of programmes aim at reducing micronutrient deficiencies. These include salt iodisation and vitamin A supplementation for children under five years of age through mass campaigns linked to national immunisation days. Vitamin A supplementation for children under five years of age and postpartum women (not later than 6 weeks after delivery) through routine health services has also begun but on a limited scale. Women are given vitamin A supplements during the postpartum period to assist both the mother and her breastfeeding children.

Disorders induced by dietary iodine deficiency constitute a major global nutrition concern. A lack of sufficient iodine can lead to goitre, hypothyroidism, impaired mental functions, retarded mental and physical development, and diminished school performance. Iodine deficiency in the foetus leads to increased rates of abortion, stillbirths, congenital anomalies, cretinism, psychomotor defects, and neonatal mortality. Iodine deficiency can be avoided by using salt that has been fortified with iodine (iodised salt).

Table 10.7 presents the percent distribution of households with salt tested for iodine content by level of iodine in salt (parts per million), percentage of households tested, and percentage of households with no salt, according to background characteristics. It shows that 90 percent of the households interviewed in the 2003 GDHS had their salt tested for iodine, while 9 percent had no salt available in the household. Fifty-nine percent of households are consuming salt that is not iodised, 13 percent of households are consuming inadequately iodised salt (<15 ppm), and only 28 percent are consuming adequately iodised salt (15+ ppm). Although Ghana has a salt iodisation law, there is need for more stringent enforcement and monitoring. The proportion of households with adequately iodised salt in rural areas (16 percent) is markedly lower than in the urban areas (44 percent). There are equally marked regional differences in adequacy of iodine levels in salt, ranging from 6 percent of households in the Northern and Upper East regions to 50 percent in the Greater Accra Region.

10.3.2 Micronutrient Intake among Children

Vitamin A is an essential micronutrient for the normal functioning of the visual system, growth and develoment, resistance to disease, and for reproduction. Vitamin A is believed to improve immunity and, hence, reduce mortality rates in children and women.

Table 10.8 shows the percentage of children under age three who consumed fruits and vegetables rich in vitamin A in the seven days preceding the survey, and the percentage of children age 6-59 months who received vitamin A supplements in the 6 months preceding the survey. It also indicates the percentage of children under five living in households that have adequately iodised salt.

Table 10.8 shows that 41 percent of children under three who live with their mothers consume fruits and vegetables rich in vitamin A. The fact that 78 percent of children 6-59 months are reported to have received a vitamin A supplement in the previous 6 months is encouraging. The data indicate that only 23 percent of children under the age of three live in households that use adequately iodised salt.

Consumption of vitamin A supplements is highest in the age group 24-35 months. There are no marked differences between the sexes on the consumption of vegetables and fruits rich in vitamin A, vitamin A supplements, or iodised salt. There is also little difference by birth order in the consumption of

Table 10.7 Iodisation of household salt

Percent distribution of households with salt tested for iodine content by level of iodine in salt (parts per million), percentage of households tested, and percentage of households with no salt, according to background characteristics, Ghana 2003

Background characteristic	Iodine content among households tested: None (0 ppm)	Inadequate (<15 ppm)	Adequate (15+ ppm)	Total	Number of households	Percentage of households tested	Percentage of households with no salt	Number of households
Residence								
Urban	43.9	12.4	43.7	100.0	2,460	85.7	12.8	2,870
Rural	69.9	13.9	16.2	100.0	3,159	93.4	5.2	3,381
Region								
Western	45.5	15.5	39.1	100.0	544	88.9	10.4	612
Central	81.1	3.7	15.1	100.0	541	92.1	7.9	587
Greater Accra	47.0	3.1	49.9	100.0	731	82.2	15.3	890
Volta	78.6	6.7	14.7	100.0	491	91.2	6.3	538
Eastern	73.7	7.7	18.5	100.0	649	88.7	8.0	732
Ashanti	37.0	26.0	36.9	100.0	1,191	90.7	9.0	1,313
Brong Ahafo	43.0	21.9	35.1	100.0	614	92.3	7.2	665
Northern	87.7	6.5	5.8	100.0	462	94.9	4.3	487
Upper East	89.7	4.4	5.9	100.0	259	92.5	5.1	280
Upper West	36.8	35.1	28.1	100.0	137	93.2	1.3	147
Wealth quintile								
Lowest	80.2	11.0	8.8	100.0	932	96.0	2.7	971
Second	73.2	13.8	13.0	100.0	1,100	94.2	4.7	1,168
Middle	63.4	16.5	20.1	100.0	1,186	90.2	7.9	1,315
Fourth	53.4	15.4	31.2	100.0	1,248	85.9	12.9	1,452
Highest	27.5	8.7	63.8	100.0	1,152	85.7	12.5	1,345
Total	58.5	13.2	28.3	100.0	5,619	89.9	8.7	6,251

ppm=parts per million

vitamin A supplements or foods rich in vitamin A. However, children of birth order greater than three are less likely than children of birth order 1-3, to be living in households that use adequately iodised salt. As expected, non-breastfeeding children consume more micronutrient-rich foods than breastfeeding children. There are minimal differences between urban and rural children in the consumption of fruits and vegetables rich in vitamin A and vitamin A supplementation. Data also indicate that the consumption of foods rich in vitamin A is higher for children of women with a secondary or higher education compared with those with primary or no education. There are marked regional differences in micronutrient intake among children. For example, the level of vitamin A supplementation among children ranges from 67 percent in the Central Region to 86 percent in the Upper East Region.

Table 10.8 Micronutrient intake among children

Percentage of youngest children under age three living with the mother who consumed fruits and vegetables rich in vitamin A in the seven days preceding the survey, percentage of children age 6-59 months who received vitamin A supplements in the six months preceding the survey, and percentage of children under five living in households using adequately iodised salt, by background characteristics, Ghana 2003

Background characteristic	Consumed fruits and vegetables rich in vitamin A[1]	Number of children under age three	Consumed vitamin A supplements	Number of children age 6-59 months	Percentage living in households using adequately iodised salt[2]	Number of children under five
Age in months						
<6	1.5	308	na	na	23.7	299
6-9	23.4	239	65.4	241	22.3	230
10-11	46.0	131	74.3	134	16.8	127
12-23	51.4	662	79.1	695	21.5	662
24-35	59.2	507	82.8	649	24.7	620
36-47	na	na	80.8	695	20.2	655
48-59	na	na	75.9	612	25.3	577
Sex						
Male	42.1	934	79.8	1,515	22.2	1,591
Female	40.3	912	76.9	1,511	23.0	1,577
Birth order						
1	41.4	391	77.0	690	27.8	715
2-3	42.5	693	78.1	1,067	26.4	1,118
4-5	39.3	376	78.1	687	17.3	711
6+	40.8	385	80.7	581	15.9	624
Breastfeeding status						
Breastfeeding	33.6	1,330	75.5	1,052	19.9	1,304
Not breastfeeding	60.5	509	79.8	1,930	24.6	1,822
Residence						
Urban	40.8	629	80.6	1,017	42.5	1,037
Rural	41.5	1,217	77.2	2,009	12.9	2,131
Region						
Western	40.5	168	80.1	297	31.5	318
Central	49.6	156	66.8	259	5.6	263
Greater Accra	34.5	194	74.3	339	40.9	328
Volta	39.8	151	82.2	242	13.6	252
Eastern	50.5	184	78.4	313	11.8	320
Ashanti	51.3	345	82.0	563	39.0	588
Brong Ahafo	47.7	206	75.1	328	32.8	353
Northern	27.2	263	78.3	409	4.1	445
Upper East	23.9	120	85.5	186	3.3	202
Upper West	34.1	60	84.8	91	26.0	99
Mother's education						
No education	33.8	744	74.5	1,200	11.8	1,295
Primary	42.5	421	75.0	694	19.0	707
Middle/JSS	47.2	574	84.1	958	31.8	992
Secondary+	56.5	107	86.2	174	64.8	175
Mother's age at birth						
<20	41.1	207	73.9	339	21.2	349
20-24	42.5	447	76.0	765	22.5	774
25-29	39.7	464	79.9	756	24.5	807
30-34	39.5	359	79.3	587	21.7	616
35-49	43.3	370	81.2	579	22.1	622
Wealth quintile						
Lowest	32.7	486	76.3	788	8.2	840
Second	48.0	398	75.6	658	12.7	718
Middle	39.6	369	79.2	576	18.9	603
Fourth	43.8	308	78.2	530	29.6	542
Highest	45.7	284	84.8	474	60.6	466
Total	41.2	1,846	78.4	3,026	22.6	3,169

Note: Information on vitamin A supplements is based on mother's recall. Figures in parentheses are based on 25-49 unweighted cases.
na = Not applicable
[1] Includes pumpkin, red or yellow yams or squash, carrots, red sweet potatoes, green leafy vegetables, mango, papaya, and other fruits and vegetables that are rich in vitamin A
[2] Salt containing 15 ppm of iodine or more. Excludes children in households in which salt was not tested.

10.3.3 Micronutrient Intake Among Mothers

It is recommended that mothers in Ghana be given two doses of vitamin A (24 hours apart) within 8 weeks of delivery. Mothers who deliver at health institutions are to receive a dose of vitamin A before they are discharged from the institution and given the second dose to take at home the following day. This is to boost the vitamin A level for their benefit and, through their breast milk, for the benefit of the child.

Table 10.9 shows the percentage of women with a birth in the five years preceding the survey who received a vitamin A dose in the first two months after delivery, the percentage who experienced night blindness during pregnancy, the percentage who took iron tablets for a specific number of days, and the percentage who live in households using adequately iodised salt, by background characteristics. The table shows that only 43 percent of mothers with a birth in the last five years reported receiving a vitamin A dose postpartum. Vitamin A supplementation is slightly higher among mothers of first births. Vitamin A supplementation among women is higher in urban areas (50 percent) than in rural areas (39 percent). The Upper East Region, where the programme was started before other regions, has the highest level of postpartum supplementation of vitamin A (73 percent), while the Western Region has the lowest (21 percent). This indicates the need for strengthening routine supplementation by health services immediately after a birth, because it is the best opportunity to capture most of the mothers.

Table 10.9 also shows that 8 percent of interviewed women reported night blindness during pregnancy. When the data are adjusted for blindness not attributed to vitamin A deficiency during pregnancy, only 2 percent of women reported night blindness during their last pregnancy. Women age 25-29 and those who have had six or more births are more likely to report a higher prevalence of night blindness associated with vitamin A deficiency during pregnancy. Night blindness is notably higher in the Upper West, Volta, and Northern regions of the country. Prevalence of night blindness is also higher among women with little or no education and women from the two lowest wealth quintiles.

Iron-deficiency anaemia is a major threat to maternal health; it contributes to low birth weight, lowered resistance to infection, poor cognitive development, and decreased work capacity. Furthermore, anaemia increases morbidity from infections because it adversely affects the body's immune response. According to the government policy, all pregnant mothers attending antenatal clinics are supposed to be given iron tablets during their pregnancy. International recommendations are that iron tablets be taken daily for at least three months during pregnancy. It should be noted that although four in five women reported having taken iron tablets during their pregnancy, only 40 percent took them for 90 or more days (Table 10.9).

According to the baseline study on prevalence and etiology of anaemia conducted in 1995 (GHS, 1995), 65 percent of pregnant women, 59 percent of lactating mothers, 71 percent of school-age children and 84 percent of preschool children were found to be anaemic. This situation poses a challenge to the Ghana government if morbidity and mortality due to anaemia and other micronutrient deficiencies is to be reduced.

Table 10.9 shows that only 24 percent of women live in households where adequately iodised salt is used. There are no marked differences in the level of micronutrient supplementation by mother's age at birth or number of children ever born.

Table 10.9 Micronutrient intake among mothers

Percentage of women with a birth in the five years preceding the survey who received a vitamin A dose in the first two months after delivery, percentage who suffered from night blindness during pregnancy, percentage who took iron tablets or syrup for specific number of days, and percentage who live in households using adequately iodised salt, by background characteristics, Ghana 2003

Background characteristic	Received vitamin A dose postpartum[1]	Night blindness during pregnancy		Number of days iron tablets taken during pregnancy						Number of women	Percentage living in households using adequately iodised salt[3]	Number of women
		Reported	Adjusted[2]	None	<60	60-89	90+	Don't know	Missing			
Age at birth												
<20	42.0	6.5	1.4	18.7	21.7	9.5	40.2	9.8	0.1	297	21.9	277
20-24	42.6	7.3	1.7	18.0	22.9	8.8	40.7	9.7	0.0	607	24.3	570
25-29	41.0	8.3	2.5	18.0	19.3	9.6	40.3	12.4	0.4	647	25.1	615
30-34	44.2	5.7	1.5	19.7	18.3	8.2	42.6	9.8	1.3	508	24.5	486
35-49	45.3	9.6	2.1	20.2	21.7	9.0	35.3	12.1	1.8	586	23.3	564
Number of children ever born												
1	47.1	6.9	1.6	16.8	19.1	11.4	42.4	10.2	0.0	565	28.6	538
2-3	41.5	7.3	1.9	17.4	20.9	6.2	42.2	12.2	1.1	940	28.5	887
4-5	42.7	6.6	1.4	19.7	19.3	8.5	41.2	10.0	1.2	582	18.9	548
6+	41.9	10.1	2.8	22.7	23.6	11.8	31.1	10.2	0.5	558	17.5	538
Residence												
Urban	49.7	5.9	2.0	12.6	12.8	6.7	54.9	12.1	0.9	946	43.8	881
Rural	39.3	8.6	1.9	22.4	25.2	10.2	31.2	10.2	0.7	1,699	13.4	1,631
Region												
Western	20.5	7.0	2.3	21.5	17.0	9.2	39.5	12.9	0.0	246	34.8	233
Central	39.7	4.6	1.3	17.0	12.0	13.4	57.2	0.5	0.0	211	6.9	201
Greater Accra	40.5	7.1	2.0	14.1	10.2	4.3	56.4	14.6	0.3	303	42.7	271
Volta	24.1	12.0	5.0	15.3	50.0	13.8	15.6	5.0	0.3	220	12.7	206
Eastern	34.3	6.5	0.5	21.4	26.8	8.9	35.2	5.0	2.7	266	13.0	251
Ashanti	50.7	4.0	0.8	14.4	14.4	13.9	53.9	2.3	1.0	507	39.5	486
Brong Ahafo	53.2	6.7	0.4	6.8	15.0	10.1	39.2	28.6	0.3	297	35.6	285
Northern	46.8	12.8	3.7	31.8	32.2	2.0	11.9	21.4	0.7	346	4.3	338
Upper East	72.9	9.7	0.6	25.6	7.6	4.5	58.0	3.4	0.9	166	3.8	161
Upper West	47.7	12.0	5.6	39.2	33.0	5.7	7.8	13.1	1.1	83	26.2	79
Education												
No education	43.4	11.0	2.4	27.5	23.9	8.1	27.9	12.0	0.7	1,025	11.9	989
Primary	39.2	5.5	1.4	15.9	21.7	8.7	43.2	9.9	0.6	589	20.1	552
Middle/JSS	44.3	5.1	1.9	11.7	18.5	10.2	48.2	10.4	1.1	879	34.1	827
Secondary+	48.0	8.2	0.7	14.5	9.1	8.9	56.7	10.8	0.0	153	65.2	143
Wealth quintile												
Lowest	40.5	10.1	2.1	29.5	24.9	7.7	25.1	12.1	0.7	648	8.0	629
Second	39.5	10.4	2.7	19.2	26.8	12.3	31.8	9.7	0.3	557	12.6	541
Middle	39.4	5.7	1.5	16.7	25.0	10.8	38.6	8.1	0.8	534	20.2	493
Fourth	48.1	6.8	1.5	14.3	13.0	8.2	49.9	14.0	0.5	474	31.0	451
Highest	50.4	3.9	1.5	10.4	10.0	5.3	61.9	10.6	1.7	433	61.9	398
Total	43.0	7.7	1.9	18.9	20.7	9.0	39.7	10.9	0.8	2,645	24.1	2,512

Note: For women with two or more live births in the five-year period, data refer to the most recent birth.
[1] In the first two months after delivery
[2] Women who reported night blindness but did not report difficulty with vision during the day
[3] Salt containing 15 ppm of iodine or more. Excludes women in households in which salt was not tested.

10.3.4 Prevalence of Anaemia in Children

The most common causes of anaemia in Ghana are inadequate dietary intake of iron, malaria, and intestinal worm infestation (GHS, 2003a). Iron and folic acid supplementation and anti-malarial prophylaxis for pregnant women, promotion of the use of insecticide-treated bed nets by pregnant women and children under five, and six-month de-worming for children age two to five years are some of the measures being pursued by the Ghana Health Service to reduce anaemia prevalence among vulnerable groups.

Table 10.10 shows the percentage of children age 6-59 months classified as having anaemia, by background characteristics. Three-quarters of Ghanaian children 6-59 months old have some level of anaemia, including 23 percent of children who are mildly anaemic, 47 percent who are moderately anaemic, and 6 percent who are severely anaemic.

Prevalence of anaemia among children is higher in rural areas (80 percent) than in urban areas (68 percent). The Northern Region has the highest prevalence of anaemia (83 percent), while the Greater Accra Region has the lowest (61 percent). Prevalence of anaemia is also higher among children of mothers with little or no education, young mothers (15-19 years), and children in households in the two poorest wealth quintiles. This indicates the widespread nature of the problem and the need to intensify the various components of the anaemia control strategy.

10.3.5 Prevalence of Anaemia in Women

Table 10.11 presents information on prevalence of anaemia among women age 15-49. The prevalence of anaemia is less pronounced among women than among children. Forty-five percent of women age 15-49 are anaemic, with 35 percent mildly anaemic, 9 percent moderately anaemic, and less than 1 percent severely anaemic. Differences by urban-rural place of residence are not large. About half of women in seven regions in Ghana (including Greater Accra) suffer from some degree of anaemia. Women residing in Brong Ahafo Region are least likely to be anaemic. Lack of education, being pregnant, and living in poor households are also associated with higher prevalence of anaemia in women of childbearing age.

Table 10.10 Prevalence of anaemia in children

Percentage of children age 6-59 months classified as having anaemia, by background characteristics, Ghana 2003

Background characteristic	Any anaemia	Anaemia status			Number of children
		Mild anaemia 10.0-10.9 g/dl)	Moderate anaemia (7.0-9.9 g/dl)	Severe anaemia (below 7.0 g/dl)	
Age in months					
6-9	74.5	19.5	46.6	8.4	227
10-11	85.7	24.0	48.6	13.0	121
12-23	83.5	19.2	54.2	10.1	661
24-35	76.4	21.3	50.2	5.0	635
36-47	73.5	24.4	45.9	3.2	716
48-59	69.5	28.1	38.8	2.5	632
Sex					
Male	76.2	22.5	47.9	5.8	1,481
Female	75.9	23.5	46.7	5.7	1,511
Birth order[1]					
1	75.4	24.1	46.0	5.2	566
2-3	76.6	23.2	48.5	4.8	935
4-5	76.4	21.0	47.5	7.9	624
6+	78.1	21.4	50.3	6.4	537
Birth interval in months[1]					
First birth[2]	75.1	24.0	45.5	5.5	573
<24	81.6	22.0	50.0	9.6	273
24-47	78.9	21.8	51.1	6.0	1,142
48+	72.0	22.8	44.3	4.9	673
Residence					
Urban	67.8	26.1	37.8	4.0	984
Rural	80.1	21.5	51.9	6.7	2,008
Region					
Western	80.4	23.8	47.6	9.0	293
Central	76.8	24.5	46.4	5.9	267
Greater Accra	61.3	24.0	33.2	4.1	324
Volta	72.7	25.3	45.6	1.8	255
Eastern	74.4	24.2	46.5	3.7	292
Ashanti	79.0	23.1	48.5	7.4	553
Brong Ahafo	74.9	22.4	45.9	6.6	333
Northern	82.5	18.7	58.1	5.7	403
Upper East	79.1	22.2	49.4	7.5	186
Upper West	78.3	23.9	52.0	2.5	86
Mother's education[3]					
No education	81.6	20.4	55.2	6.0	1,097
Primary	77.9	23.6	47.3	7.0	625
Middle/JSS	71.8	24.0	41.7	6.0	859
Secondary+	60.6	27.9	31.8	0.8	154
Mother's age[3]					
15-19	86.5	29.5	47.3	9.7	76
20-24	79.1	17.9	53.0	8.2	490
25-29	77.5	22.8	49.7	5.0	706
30-34	75.1	25.8	44.5	4.9	607
35-49	74.3	22.5	46.0	5.8	857
Mother's status					
Mother interviewed	76.6	22.5	48.1	6.0	2,660
Mother not interviewed but in the household	73.5	27.4	41.5	4.6	75
Mother not in the household[4]	71.2	26.4	40.7	4.1	257
Wealth quintile					
Lowest	82.2	19.2	55.5	7.5	774
Second	81.7	23.0	51.7	7.1	660
Middle	79.2	23.4	50.0	5.8	597
Fourth	68.8	23.5	40.7	4.7	521
Highest	61.1	28.6	30.4	2.1	441
Total	76.1	23.0	47.3	5.8	2,992

Note: Table is based on children who stayed in the household the night before the interview.
g/dl = grams per deciliter
[1] Excludes children whose mothers were not interviewed
[2] First born twins (triplets, etc.) are counted as first births because they do not have a previous birth interval
[3] For women who are not interviewed, information is taken from the Household Questionnaire. Excludes children whose mothers are not listed in the household schedules.
[4] Includes children whose mothers are deceased

Table 10.11 Prevalence of anaemia in women

Percentage of women age 15-49 with anaemia, by background characteristics, Ghana 2003

Background characteristic	Any anaemia	Anaemia status			Number of women
		Mild anaemia	Moderate anaemia	Severe anaemia	
Age[1]					
15-19	45.8	37.2	7.9	0.7	1,060
20-24	45.0	34.1	10.2	0.7	939
25-29	42.5	34.1	8.1	0.2	893
30-34	43.8	35.9	7.1	0.8	729
35-39	47.5	36.1	9.8	1.6	669
40-44	43.9	32.4	10.9	0.6	529
45-49	44.5	32.4	10.5	1.6	454
Children ever born[2]					
None	43.6	34.1	8.6	0.8	1,635
1	45.3	34.6	9.6	1.1	740
2-3	43.6	34.8	8.4	0.4	1,223
4-5	47.2	37.0	9.7	0.5	851
6+	45.6	34.7	9.4	1.4	823
Maternity status[2]					
Pregnant	64.9	37.0	26.7	1.2	400
Breastfeeding	47.9	40.1	7.3	0.5	1,262
Neither	41.4	32.9	7.6	0.9	3,610
Using IUD[2]					
Yes	31.6	27.5	4.2	0.0	31
No	44.8	35.0	9.0	0.8	5,242
Residence					
Urban	41.6	32.4	8.2	1.1	2,524
Rural	47.6	37.2	9.7	0.6	2,748
Region					
Western	39.1	31.9	6.3	0.8	531
Central	38.0	29.4	7.5	1.1	414
Greater Accra	47.0	37.8	8.5	0.7	861
Volta	48.5	36.0	12.2	0.3	468
Eastern	48.4	37.4	10.0	1.0	524
Ashanti	46.2	34.3	10.5	1.4	1,078
Brong Ahafo	33.5	26.3	6.9	0.2	544
Northern	49.8	42.7	6.4	0.6	445
Upper East	51.0	35.0	15.4	0.6	272
Upper West	49.7	44.9	4.5	0.3	136
Education[1]					
No education	48.7	37.9	10.0	0.8	1,485
Primary	44.5	33.9	10.0	0.7	1,059
Middle/JSS	44.5	35.2	8.4	0.9	2,122
Secondary +	36.1	28.4	6.8	0.8	606
Wealth quintile					
Lowest	53.7	42.1	11.1	0.5	899
Second	45.6	35.0	10.1	0.6	893
Middle	47.3	36.4	9.7	1.2	1,012
Fourth	40.7	31.7	8.4	0.5	1,153
Highest	39.6	31.5	6.8	1.2	1,316
Total	44.7	34.9	9.0	0.8	5,272

Note: Table is based on women who stayed in the household the night before the interview. Women with <7.0 g/dl of hemoglobin have severe anaemia, women with 7.0-9.9 g/dl have moderate anaemia, and pregnant women with 10.0-10.9 g/dl and nonpregnant women with 10.0-11.9 g/dl have mild anaemia.
g/dl = grams per deciliter
[1] For women who are not interviewed, information is taken from the Household Questionnaire
[2] Excludes women who were not interviewed

10.3.6 Prevalence of Anaemia in Children by Anaemia Status of Mother

Table 10.12 shows the percentage of children age 6-59 months classified as anaemic, by the anaemia status of the mother. There is not a strong relationship between the anaemia status of mothers and the anaemia status of children except in the case of mothers with moderate anaemia, whose children are more likely to suffer from severe anaemia (9 percent) than children of mothers with no anaemia (5 percent) or those with mild anaemia (6 percent).

Table 10.12 Prevalence of anaemia in children by anaemia status of mother

Percentage of children age 6-59 months classified as having anaemia, by anaemia status of mother, Ghana 2003

		Anaemia status of child			
Anaemia status of mother	Any anaemia	Mild (10.0-10.9 g/dl)	Moderate (7.0-9.9 g/dl)	Severe (below 7.0 g/dl)	Number of children
No anaemia	73.3	23.0	45.0	5.3	1,391
Any anaemia	80.6	22.1	51.7	6.7	1,237
Mild anaemia	80.0	21.5	52.3	6.1	988
Moderate anaemia	83.1	24.2	50.0	8.8	239
Severe anaemia	*	*	*	*	10
Total	76.7	22.6	48.2	6.0	2,629

Note: Table is based on children who stayed in the household the night before the interview. Table includes only cases with anaemia measurements for both mothers and children. An asterisk indicates that a figure is based on fewer than 25 unweighted cases and has been suppressed.
g/dl = grams per deciliter

10.4 NUTRITIONAL STATUS OF CHILDREN UNDER AGE FIVE

The 2003 GDHS included information on the nutritional status of children under five years of age for three indices, namely, weight-for-age, height-for-age, and weight-for-height.

10.4.1 Measures of Nutritional Status in Childhood

In addition to questions on feeding practices associated with infant and young children, the 2003 GDHS included anthropometric measurements for all children under five years of age. Each interviewing team was equipped with a lightweight electronic SECA scale, designed and manufactured under the guidance of UNICEF, and a measuring board specifically produced by Shorr Productions for use in survey settings. Children younger than 24 months were measured lying down on the board (recumbent length), while standing height was measured for older children. The scale also allowed for the weighing of very young children, with an automatic mother-child adjustment, which eliminated the mother's weight while she stood with her baby on the weighing scale.

In previous GDHS surveys, anthropometric measurements were restricted to children born to women interviewed with the Women's Questionnaire. However, these data are not representative of all children, since they exclude children whose mothers were not in the household (either because they did not live there, or because they had died), children whose mothers were not eligible for the individual interview (i.e., under age 15 or age 50 and over), and children whose mothers did not complete an

individual interview. To overcome these biases, the 2003 GDHS included weight and height measurements of all children who were born in the five years preceding the survey and listed in the Household Questionnaire, irrespective of the interview status of their mother.

As recommended by the World Health Organisation (WHO), the anthropometric measurements of children in the survey are compared with an international reference population defined by the U.S. National Centre for Health Statistics (NCHS) and accepted by the U.S. Centres for Disease Control and Prevention (CDC). Each of the three nutritional status indicators described below are expressed in standard deviation units (Z-scores) from the median for the reference population. The use of this reference population is based on the finding that well nourished young children of all population groups (for which data exist) follow very similar growth patterns. The reference populations serve as a point of comparison, facilitating the examination of differences in the anthropometric status of subgroups in a population and changes in nutritional status over time. In any large population, there is variation in height and weight; this variation approximates a normal distribution.

Each of these indices—height-for-age, weight-for-height, and weight-for-age—gives different information about growth and body composition used to assess nutritional status. The height-for-age index is an indicator of linear growth retardation. Children whose height-for-age Z-score is below minus two standard deviations (-2 SD) from the median of the reference population are considered short for their age (stunted) and are chronically malnourished. Children who are below minus three standard deviations (-3 SD) from the median of the reference population are considered severely stunted. Stunting reflects failure to receive adequate nutrition over a long period of time and is also affected by recurrent and chronic illness. Height-for-age, therefore, represents the long-term effect of malnutrition in a population and does not vary according to recent dietary intake.

The weight-for-height index measures body mass in relation to body length and describes current nutritional status. Children whose Z-scores are below minus two standard deviations (-2 SD) from the median of the reference population are considered thin (wasted) for their height and are acutely malnourished. Wasting represents the failure to receive adequate nutrition in the period immediately preceding the survey and may be the result of inadequate food intake or a recent episode of illness causing loss of weight and the onset of malnutrition. Children whose weight-for-height is below minus three standard deviations (-3 SD) from the median of the reference population are considered severely wasted.

Weight-for-age is a composite index of height-for-age and weight-for-height. It takes into account both acute and chronic malnutrition. Children whose weight-for-age is below minus two standard deviations from the median of the reference population are classified as underweight.

Table 10.13 shows the percentage of children under five years classified as malnourished according to height-for-age, weight-for-height, and weight-for-age, by background characteristics. A total of 3,586 children under five were identified in the households. Eight percent of children had missing information on height or weight, 3 percent had implausibly high or low values for height or weight measures, and less than 1 percent had incomplete age information. The analysis on nutritional status is based on the remaining 89 percent or 3,183 children.

According to the 2003 GDHS, 30 percent of children under five are stunted and 11 percent severely stunted. Seven percent of children under five are wasted and 1 percent severely wasted. Weight-for-age results show that 22 percent of children under five are underweight, with 5 percent severely underweight.

Table 10.13 and Figure 10.3 indicate that stunting is evident among children as young as 6 months of age (6 percent). Stunting increases with the age of the child. This is evidenced by the marked increase in stunting levels from 14 percent at 6-9 months to 35 percent at age 12-23 months. There are no

marked differences between males (33 percent) and females (27 percent) in the levels of stunting among children under five years. Table 10.13 also shows that the longer the birth interval, the lower the stunting levels. Birth size is an important indicator of a child's nutritional status at birth and in the future. According to Table 10.13, a larger percentage of children who were reported to have been small or very small at birth were stunted (40 percent) compared with children who were average or larger in size at birth (27 percent).

Table 10.13 indicates that wasting is more common in the age group 6-23 months and decreases as the child ages. Weight-for-age is an index of chronic or acute malnutrition. It does not distinguish between a child who is underweight because of stunting or wasting. Underweight is very minimal for children less than 6 months of age but becomes more pronounced at age 6 months and above (during the normal complementary feeding period). This may be because most children are still primarily breastfeeding until 6 months of age.

Table 10.13 also shows that rural children are more likely to be stunted and underweight than urban children, while the proportion wasted is almost the same in both rural and urban areas (7 percent). Regional variation in nutritional status of children is substantial. The Central, Northern, Upper East and Upper West regions have stunting levels that are above the national average. Children living in the Upper East, Northern, and Upper West regions also have underweight levels above the national average, in addition to children from the Volta Region. Size at birth is also related to wasting in children; a larger percentage of children reported by mothers as very small at birth are wasted (12 percent), compared with those reported as average or larger (7 percent). Wasting is particularly high in the Volta, Upper East, and Upper West regions.

The percentage of malnourished children decreases with increasing level of mother's education. About a third of children whose mothers have no education are stunted or underweight. The pattern is less clear with levels of wasting. Mother's age has no marked influence on the nutritional status of the child. There are no strong differences by sex and birth order.

Table 10.13 indicates that children whose biological mothers were not in the household are more likely to be malnourished (34 percent stunted and 25 percent underweight) than children whose mothers were interviewed.

Poor nutritional status among children is obvious among those who live in poor households. For example, two-fifths of children who live in households in the lowest wealth quintile are stunted compared with one-eighth of children in the highest wealth quintile. A similar pattern is seen among children who are underweight. The relationship between wasting and wealth is not that strong.

Table 10.13 Nutritional status of children

Percentage of children under five years classified as malnourished according to three anthropometric indices of nutritional status: height-for-age, weight-for-height, and weight-for-age, by background characteristics, Ghana 2003

Background characteristic	Height-for-age			Weight-for-height			Weight-for-age			Number of children
	Percentage below -3 SD	Percentage below -2 SD[1]	Mean Z-score SD	Percentage below -3 SD	Percentage below -2 SD[1]	Mean Z-score SD	Percentage below -3 SD	Percentage below -2 SD[1]	Mean Z-score SD	
Age in months										
<6	1.0	6.3	0.0	2.0	5.9	0.1	0.0	2.4	0.2	248
6-9	3.0	14.3	(0.3)	3.5	13.5	(0.7)	2.9	13.6	(0.8)	226
10-11	3.6	17.1	(0.9)	3.2	16.3	(0.8)	8.2	35.4	(1.4)	117
12-23	12.3	35.2	(1.6)	1.7	12.6	(0.7)	7.9	30.8	(1.4)	636
24-35	12.8	32.5	(1.5)	1.0	5.5	(0.5)	7.0	26.4	(1.3)	636
36-47	13.3	35.0	(1.5)	0.8	3.8	(0.2)	4.1	21.3	(1.1)	710
48-59	12.5	33.4	(1.5)	0.7	3.4	(0.2)	1.6	18.2	(1.1)	611
Sex										
Male	12.5	32.8	(1.4)	1.7	7.2	(0.4)	4.9	22.6	(1.1)	1,588
Female	9.2	27.0	(1.2)	1.0	7.1	(0.4)	4.5	21.6	(1.1)	1,595
Birth order[2]										
1	9.4	27.2	(1.2)	1.5	7.3	(0.4)	4.4	21.1	(1.1)	613
2-3	10.2	26.5	(1.2)	1.4	7.0	(0.4)	4.2	21.0	(1.1)	1,019
4-5	11.8	32.3	(1.4)	1.4	7.3	(0.4)	4.8	24.3	(1.1)	660
6+	10.9	34.1	(1.4)	1.0	6.9	(0.4)	5.0	21.5	(1.1)	590
Birth interval in months[2]										
First birth[3]	9.4	27.0	(1.2)	1.5	7.4	(0.4)	4.4	21.1	(1.1)	620
<24	16.9	36.8	(1.6)	1.2	5.1	(0.3)	4.7	23.5	(1.2)	291
24-47	12.1	33.1	(1.4)	1.5	7.5	(0.4)	5.3	24.0	(1.2)	1,245
48+	6.3	22.8	(1.1)	1.1	6.9	(0.3)	3.5	18.2	(0.9)	726
Size at birth[2]										
Very small	13.8	39.8	(1.6)	0.7	12.0	(0.6)	8.9	35.4	(1.5)	179
Small	14.8	39.7	(1.6)	1.4	6.2	(0.6)	6.7	31.5	(1.4)	337
Average or larger	9.8	27.4	(1.2)	1.4	6.8	(0.3)	4.0	19.5	(1.0)	2,348
Residence										
Urban	6.8	20.5	(0.9)	1.9	6.6	(0.4)	3.2	15.4	(0.9)	1,050
Rural	12.8	34.5	(1.5)	1.1	7.4	(0.4)	5.4	25.4	(1.2)	2,132
Region										
Western	8.0	28.4	(1.3)	1.1	5.3	(0.3)	2.2	16.5	(1.0)	333
Central	12.5	31.6	(1.4)	0.0	3.0	(0.3)	4.0	22.0	(1.0)	284
Greater Accra	5.5	13.9	(0.6)	2.7	7.2	(0.4)	2.1	11.5	(0.7)	337
Volta	7.8	23.3	(1.1)	3.1	13.9	(0.7)	4.9	25.7	(1.2)	259
Eastern	6.2	27.4	(1.2)	0.7	6.2	(0.3)	3.7	17.3	(1.0)	333
Ashanti	10.2	29.1	(1.3)	0.8	6.7	(0.4)	4.1	20.8	(1.1)	613
Brong Ahafo	10.7	29.4	(1.3)	1.3	5.7	(0.4)	5.5	20.4	(1.1)	356
Northern	21.8	48.8	(1.9)	1.0	6.6	(0.3)	8.7	35.5	(1.5)	415
Upper East	12.1	31.7	(1.3)	2.4	12.9	(0.8)	7.8	32.4	(1.4)	156
Upper West	12.6	34.1	(1.3)	2.6	11.0	(0.3)	6.0	25.9	(1.0)	95
Mother's education[4]										
No education	14.7	38.2	(1.5)	1.4	7.5	(0.4)	6.8	27.5	(1.3)	1,177
Primary	8.2	24.1	(1.1)	1.0	7.8	(0.4)	3.3	18.6	(1.0)	661
Middle/JSS	8.4	25.8	(1.2)	1.5	6.7	(0.4)	3.2	19.7	(1.0)	938
Secondary +	1.8	11.1	(0.5)	1.6	4.6	(0.3)	0.8	7.8	(0.6)	170
Mother's age										
15-19	12.4	30.1	(1.2)	1.4	11.3	(0.5)	4.5	26.4	(1.1)	93
20-24	11.4	28.9	(1.3)	2.1	8.0	(0.5)	5.1	23.6	(1.1)	530
25-29	9.4	28.2	(1.2)	1.2	6.7	(0.4)	4.0	20.6	(1.1)	779
30-34	9.9	28.0	(1.2)	1.2	8.2	(0.4)	4.3	22.7	(1.1)	645
35-49	11.1	32.1	(1.3)	1.1	5.9	(0.3)	4.8	20.9	(1.1)	900
Mother's status										
Mother interviewed	10.5	29.5	(1.3)	1.4	7.1	(0.4)	4.6	21.9	(1.1)	2,882
Mother not interviewed but in household	7.3	30.0	(1.1)	1.1	9.3	(0.6)	2.1	21.3	(1.1)	65
Mother not in household[5]	15.0	34.0	(1.5)	1.3	7.0	(0.5)	7.0	25.1	(1.3)	236
Wealth quintile										
Lowest	16.5	41.8	(1.7)	1.3	8.8	(0.5)	8.7	31.3	(1.4)	799
Second	12.3	31.5	(1.4)	1.1	6.7	(0.4)	4.8	23.9	(1.2)	716
Middle	10.1	30.2	(1.4)	1.3	6.5	(0.4)	2.9	21.5	(1.1)	655
Fourth	6.9	24.2	(1.0)	1.8	6.8	(0.4)	3.6	16.3	(1.0)	541
Highest	4.5	13.2	(0.6)	1.3	6.1	(0.3)	1.5	11.4	(0.7)	471
Total	10.8	29.9	(1.3)	1.3	7.1	(0.4)	4.7	22.1	(1.1)	3,183

Note: Table is based on children who stayed in the household the night before the interview. Each of the indices is expressed in standard deviation units (SD) from the median of the NCHS/CDC/WHO International Reference Population. The percentage of children who are more than three or more than two standard deviations below the median of the International Reference Population (-3 SD and -2 SD) are shown according to background characteristics. Table is based on children with valid dates of birth (month and year) and valid measurement of both height and weight. Total includes 2 children with missing information on size at birth.

[1] Includes children who are below -3 standard deviations (SD) from the International Reference Population median

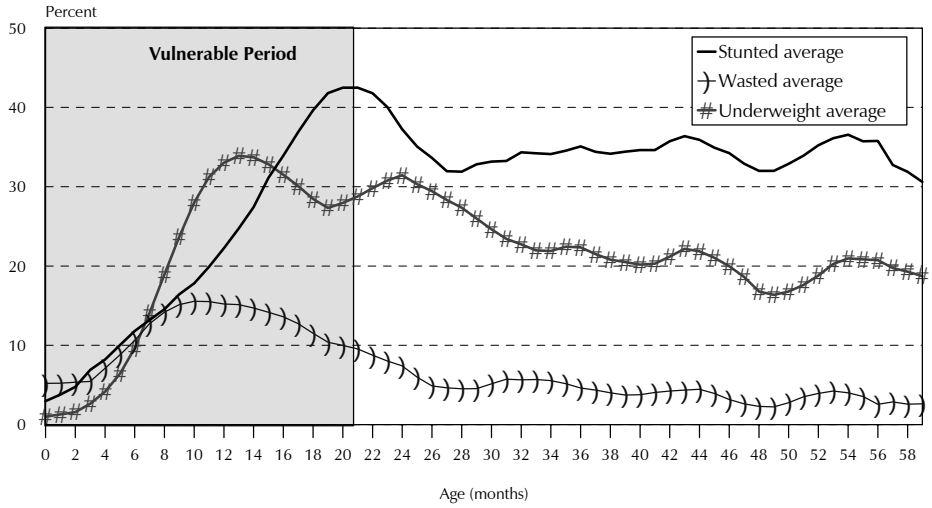
[2] Excludes children whose mothers were not interviewed

[3] First born twins (triplets, etc.) are counted as first births because they do not have a previous birth interval

[4] For women who are not interviewed, information is taken from the Household Questionnaire. Excludes children whose mothers are not listed in the household schedule.

[5] Includes children whose mothers are deceased

Figure 10.3
Stunting, Wasting, and Underweight by Age, Ghana



Note: *Stunting* reflects chronic malnutrition; *wasting* reflects acute malnutrition; *underweight* reflects chronic or acute malnutrition or a combination of both. Plotted values are smoothed by a five-month moving average.

GDHS 2003

10.4.2 Trends in Children's Nutritional Status

An analysis of trends in these data shows an increase in the level of stunting but slight decreases in the proportions of children who are wasted or underweight. Caution has to be exercised when comparing data from the various DHS surveys conducted in Ghana to assess the trend in the nutritional status. This is because information on children's nutritional status was gathered for different age groups from one survey to another. In addition, previous DHS surveys collected anthropometric measurements only for children of interviewed women, whereas the 2003 survey collected nutritional status information from all children under age five living in the household at the time of the interview. While it is possible to adjust for some of these inconsistencies, it is not possible to correct them all. Comparisons are therefore restricted to the last five years.

The proportion of children under five who are stunted has increased from 26 percent in 1998 to 30 percent in 2003. The proportion underweight decreased from 10 percent in 1998 to 7 percent in 2003. The proportion of children who are wasted also decreased from 25 percent in 1998 to 22 percent in 2003 (GSS and MI, 1999). These trends in nutritional status imply that the problem of malnutrition in Ghana may be due to chronic food shortages. This is evidenced by the fact that the three northern regions, which have annual periods of drought, have consistently recorded the highest levels of stunting.

10.5 NUTRITIONAL STATUS OF WOMEN

The 2003 GDHS collected information on the height and weight of women. The data were used to derive two measures of nutritional status, height and body mass index (BMI). A woman's height may be used to predict the risk of difficulty in delivery (given the relationship between height and the size of the pelvis). The risk of giving birth to a low-birth-weight baby is influenced by the mother's nutritional status. The cut-off point for height at which mothers can be considered at risk varies between populations but normally falls between 140 centimetres and 150 centimetres; a cutoff point of 145 centimetres is used here. The index used to measure thinness or obesity is known as the body mass index (BMI), or the

Quetelet index. BMI is defined as weight in kilogrammes divided by height squared in metres (kg/m^2). A cut-off point of 18.5 is used to define thinness or acute undernutrition. A BMI of 25 or above usually indicates overweight or obesity.

The 2003 GDHS collected anthropometric information for all women age 15-49, in contrast to earlier surveys, which restricted anthropometric data collection to women who had a live birth in the five years (or three years) preceding the survey. As such, this is a more representative sample because it includes both younger women who may not yet have given birth and older women who would have stopped childbearing. Women for whom there was no information on height and/or weight and for whom a BMI could not be estimated are excluded from this analysis.

Table 10.14 shows the nutritional status of women in the reproductive ages 15-49. The mean height of women is 159 centimetres, which is above the critical height of 145 centimetres. Nine percent of women are found to be chronically malnourished (BMI less than 18.5), while 25 percent are overweight. Variations between urban and rural women are marked. More women have a BMI less than 18.5 in rural areas (12 percent) than in urban areas (6 percent). The percentage of overweight or obese women is, however, higher in urban areas (35 percent) than in rural areas (16 percent). The Greater Accra Region has the highest percentage of these women (46 percent) and the Upper West Region has the lowest percentage (7 percent). The percentage of overweight or obese women increases with increasing educational level as well as increasing wealth quintile.

Interpretation of trends in women's nutritional status is complicated by the fact that in the 1993 and the 1998 GDHS only mothers of children under five were measured. In 2003, the data refer to all women age 15-49. A comparison, restricting the data for 2003 to mothers of children under five years, shows that there has been little change in the percentage of mothers whose height is below 145 centimetres and in the mean BMI over the past ten years, from 1993 to 2003. However, there has been a small decline in the percentage of mothers who fall below a BMI of 18.5, from 11 percent in 1993 and 1998 to 9 percent in 2003.

Table 10.14 Nutritional status of women by background characteristics

Among women age 15-49, mean height, percentage under 145 cm, mean body mass index (BMI), and percentage with specific BMI levels, by background characteristics, Ghana 2003

	Height			BMI [1] (kg/m^2)									
Background characteristic	Mean height in cm	Percentage below 145 cm	Number of women	Mean BMI	18.5-24.9 (normal)	<18.5 (thin)	17.0-18.4 (mildly thin)	16.0-16.9 (moderately thin)	<16.0 (severely thin)	≥25.0 (overweight or obese)	25.0-29.9 (overweight)	≥30.0 (obese)	Number of women
Age													
15-19	157.7	2.0	1,075	21.1	76.8	15.2	10.9	2.8	1.5	8.0	6.9	1.1	1,002
20-24	158.4	1.5	952	22.4	75.6	7.5	5.3	1.7	0.5	16.9	14.7	2.2	830
25-29	159.2	0.8	909	23.0	66.4	8.3	6.7	1.5	0.1	25.3	18.8	6.4	771
30-34	159.5	1.1	762	24.1	59.5	7.1	6.0	0.7	0.4	33.3	21.7	11.7	668
35-39	159.5	0.7	692	24.4	55.2	7.0	5.8	0.9	0.4	37.8	23.9	13.9	611
40-44	159.1	0.9	545	24.8	53.2	7.1	6.0	0.9	0.3	39.7	23.0	16.7	515
45-49	159.2	0.3	451	24.1	56.3	9.7	6.1	2.4	1.2	34.1	20.0	14.0	439
Residence													
Urban	159.2	1.1	2,556	24.2	58.4	6.4	5.4	0.7	0.3	35.1	22.4	12.7	2,378
Rural	158.5	1.2	2,831	22.0	72.2	12.0	8.5	2.5	1.0	15.8	12.2	3.6	2,457
Region													
Western	158.5	1.8	542	22.9	63.0	11.2	9.0	1.5	0.7	25.8	18.8	6.9	498
Central	157.8	2.3	423	23.2	67.1	6.7	4.4	2.0	0.3	26.2	20.5	5.7	380
Greater Accra	159.6	0.7	898	25.3	49.4	4.3	3.9	0.3	0.1	46.3	28.8	17.4	848
Volta	157.6	2.4	475	22.6	71.5	9.7	7.0	2.1	0.6	18.8	12.4	6.4	439
Eastern	158.6	0.7	577	23.3	66.2	8.9	6.6	1.6	0.7	24.8	17.0	7.8	530
Ashanti	158.7	1.3	1,030	23.2	64.3	9.6	7.3	1.6	0.8	26.1	17.3	8.7	917
Brong Ahafo	159.1	0.7	553	22.7	73.0	7.2	4.9	1.4	0.9	19.8	13.7	6.0	485
Northern	159.3	0.5	465	21.1	79.7	12.8	8.9	3.4	0.5	7.5	6.4	1.1	375
Upper East	159.3	0.4	280	20.5	69.5	23.1	17.8	2.6	2.7	7.4	6.9	0.6	239
Upper West	160.2	0.0	142	21.4	82.2	11.3	8.6	1.6	1.1	6.5	5.1	1.4	126
Education													
No education	158.7	0.8	1,526	22.1	71.5	12.1	8.6	2.2	1.2	16.5	11.8	4.6	1,297
Primary	158.1	2.0	1,096	23.0	65.4	10.3	7.4	2.0	0.8	24.3	17.4	6.9	983
Middle/JSS	158.8	1.1	2,151	23.4	64.5	8.2	6.6	1.2	0.4	27.3	18.1	9.2	1,977
Secondary+	160.5	0.6	615	24.7	55.2	5.1	3.8	1.1	0.2	39.7	25.9	13.8	578
Wealth quintile													
Lowest	158.7	1.0	931	21.1	76.7	15.5	11.0	3.1	1.4	7.7	6.4	1.3	794
Second	158.0	1.8	915	21.8	75.5	11.7	8.8	1.9	1.0	12.8	10.8	2.0	805
Middle	158.4	1.0	1,036	22.3	70.4	11.3	8.2	2.3	0.8	18.4	14.2	4.2	906
Fourth	158.7	1.6	1,166	23.6	63.4	6.8	5.5	1.0	0.4	29.8	20.6	9.1	1,062
Highest	159.8	0.6	1,338	25.3	50.2	4.4	3.7	0.5	0.2	45.4	27.3	18.0	1,267
Total	158.8	1.2	5,387	23.1	65.4	9.3	7.0	1.6	0.7	25.3	17.2	8.1	4,835

[1] Excludes pregnant women and women with a birth in the preceding 2 months